NEWTON'S INTERPOLATION FORMULA AND SUMS OF POWERS

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Abstract. In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the forward Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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1. Auxiliary Lemmas

This section introduces several lemmas, definitions, and conventions that we use in the main results. We use the notation for multifold sums introduced by Donald Knuth in [1].

Definition 1.1 (Multifold Sums of Powers)**.** *For non-negative integers r, n, and m, we define* multifold sums of powers *by*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

We use the following notation for the *falling factorial*.

Definition 1.2 (Falling Factorials)**.** *For integers n and k,*

$$(n)_k = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ \prod_{j=0}^{k-1}(n-j), & \text{if } k > 0. \end{cases}$$

J. F. Steffensen discusses generalized factorials in [2].

Lemma 1.3 (Forward Newton's formula)**.** *Let $f\colon \mathbb{Z} \to \mathbb{C}$ be a function and let $t \in \mathbb{Z}$. For every $n \in \mathbb{Z}$,*

$$f(n) = \sum_{k=0}^{\infty} \frac{(n-t)_k}{k!} \Delta^k f(t) = \sum_{k=0}^{\infty} \binom{n-t}{k} \Delta^k f(t),$$

where the k-th forward difference $\Delta^k f(t)$ *of f evaluated at t is defined by*

$$\Delta^k f(t) = \sum_{j=0}^{k} (-1)^{k-j} \tbinom{k}{j} f(t+j).$$

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Proof. Newton introduced this formula in [3]. Steffensen expressed it in modern notation in [4, Section 2, Equation (19)]. The identity $\frac{(n-t)_k}{k!} = \binom{n-t}{k}$ gives the second expression. □

The power case follows directly from the preceding formula.

Lemma 1.4 (Forward Newton's formula for powers)**.** *For non-negative integers n and m and an arbitrary integer t,*

$$n^m = \sum_{k=0}^{m} \binom{n-t}{k} \Delta^k t^m.$$

Proof. This is the special case of Lemma (1.3) obtained by taking $f(n) = n^m$. □

We next recall a generalized hockey-stick identity.

Lemma 1.5 (Generalized hockey-stick identity)**.** *For arbitrary integers a, b, and j,*

$$\sum_{k=a}^{b} \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have $\sum_{k=a}^{b} \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^{b} \binom{k}{j}\right) - \left(\sum_{k=0}^{a-1} \binom{k}{j}\right)$. The hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ therefore gives

$$\sum_{k=a}^{b} \binom{k}{j} = \left(\sum_{k=0}^{b} \binom{k}{j}\right) - \left(\sum_{k=0}^{a-1} \binom{k}{j}\right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This proves the identity. □

The following specialization will be useful in deriving the main results.

Lemma 1.6 (Forward hockey-stick identity)**.** *For an integer $n \geq 1$ and arbitrary integers t, k, and r,*

$$\sum_{j=1}^{n} \binom{j-t+r}{k+r} = \binom{n-t+r+1}{k+r+1} - \binom{1-t+r}{k+r+1}.$$

Proof. Set $a = 1 - t + r$ and $b = n - t + 1$ in Lemma (1.5). □

We also record a formula for multifold sums of zero powers.

Lemma 1.7 (Multifold sums of zero powers)**.** *For non-negative integers r and n,*

$$\Sigma^r\, n^0 = \binom{r+n-1}{r}.$$

Proof. (1) When $r = 0$, the definition gives $\Sigma^0\, n^0 = 1$.

(2) When $r = 1$, $\Sigma^1\, n^0 = \sum_{k=1}^{n} 1 = \binom{n}{1}$.

(3) When $r = 2$, $\Sigma^2\, n^0 = \sum_{k=1}^{n} \binom{k}{1} = \binom{n+1}{2}$.

(4) When $r = 3$, $\Sigma^3\, n^0 = \sum_{k=1}^{n} \binom{k+1}{2} = \binom{n+2}{3}$.

(5) For general r, the recurrence and hockey-stick identity give

$$\begin{aligned}\Sigma^r\, n^0 &= \sum_{k=1}^{n} \Sigma^{r-1}\, k^0 = \sum_{k=1}^{n} \tbinom{k+r-2}{r-1} \\ &= \tbinom{r+n-1}{r}.\end{aligned}$$

This proves the formula. □

The next lemma gives the upper-index negation identity for binomial coefficients.

Lemma 1.8 (Upper Binomial negation)**.** *For integers n and k,*

$$\binom{-n}{k} = (-1)^k \binom{k+n-1}{k}.$$

Proof. This identity is given in [5, p. 164, Equation (5.14)]. □

Finally, we adopt the *zero-power convention*.

Convention 1.9 (Zero Powers)**.** *For every x,*

$$x^0 = 1.$$

Donald Knuth discusses the convention $x^0 = 1$, including the case $x = 0$, in *Concrete Mathematics* [5, p. 162] and in [6].

2. Introduction

This manuscript derives formulas for sums of powers by combining the forward Newton interpolation formula for powers in Lemma (1.4) with the hockey-stick identity in Lemma (1.6). We implement the algorithm for finding closed forms for sums of powers of integers proposed in [7, Algorithm (11.2)]. More precisely, the main idea is to express Newton's formula in binomial bases associated with different difference operators. The following formulas illustrate these representations.

$$\begin{cases} f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^{[k]}}{k!} \delta^k f(a) = \sum_{k=0}^{\infty} \frac{x-a}{k} \binom{x-a+\frac{k}{2}-1}{k-1} \delta^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)_k}{k!} \Delta^k f(a) = \sum_{k=0}^{\infty} \binom{x-a}{k} \Delta^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^{(k)}}{k!} \nabla^k f(a) = \sum_{k=0}^{\infty} \binom{x-a+k-1}{k} \nabla^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!} \frac{\mathrm{d}^k f(a)}{\mathrm{d}x^k}. \end{cases}$$

Each variation of Newton's formula above has its own binomial basis:

$$\begin{cases} \frac{(x)_n}{n!} &= \frac{1}{n!} x(x-1)(x-2)\cdots(x-n+1) = \binom{x}{n}, \\ \frac{x^{(n)}}{n!} &= \frac{1}{n!} x(x+1)(x+2)\cdots(x+n-1) = \binom{x+n-1}{n}, \\ \frac{x^{[n]}}{n!} &= \frac{1}{n!} \left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = \frac{x}{n}\binom{x+\frac{n}{2}-1}{n-1}. \end{cases}$$

These representations allow us to combine Newton's formula with hockey-stick identities to derive closed forms for sums of powers of integers. This manuscript focuses on these closed forms, using forward Newton's formula and the hockey-stick identity in Lemma (1.6).

3. Main Results

We begin with Lemma (1.4), which gives forward Newton interpolation for powers. The following example illustrates the formula.

$$n^3 = 0\tbinom{n-0}{0} + 1\tbinom{n-0}{1} + 6\tbinom{n-0}{2} + 6\tbinom{n-0}{3},$$
$$n^3 = 1\tbinom{n-1}{0} + 7\tbinom{n-1}{1} + 12\tbinom{n-1}{2} + 6\tbinom{n-1}{3},$$
$$n^3 = 8\tbinom{n-2}{0} + 19\tbinom{n-2}{1} + 18\tbinom{n-2}{2} + 6\tbinom{n-2}{3},$$
$$n^3 = 27\tbinom{n-3}{0} + 37\tbinom{n-3}{1} + 24\tbinom{n-3}{2} + 6\tbinom{n-3}{3}.$$

We now derive a formula for sums of powers.

$$\Sigma^1 n^m = \sum_{k=0}^{m} \left[\sum_{j=1}^{n} \binom{j-t}{k} \right] \Delta^k t^m.$$

Setting $r = 0$ in Lemma (1.6) gives the following closed form for ordinary sums of powers.

Proposition 3.1 (Ordinary sums of powers)**.** *For non-negative integers n and m and an arbitrary integer t,*

$$\Sigma^1 n^m = \sum_{k=0}^{m} \left[\binom{n-t+1}{k+1} - \binom{1-t}{k+1} \right] \Delta^k t^m.$$

The basis $\binom{j-t+r}{k+r}$ in Lemma (1.6) is useful because it yields closed forms for sums of powers by varying the parameter r. The next example illustrates the case $r = 1$.

Example 3.2. *For integer $0 \le t \le 3$, we have*

$$\Sigma^1 n^3 = 0\left[\tbinom{n+1}{1} - \tbinom{1}{1}\right] + 1\left[\tbinom{n+1}{2} - \tbinom{1}{2}\right] + 6\left[\tbinom{n+1}{3} - \tbinom{1}{3}\right] + 6\left[\tbinom{n+1}{4} - \tbinom{1}{4}\right],$$
$$\Sigma^1 n^3 = 1\left[\tbinom{n}{2} - \tbinom{0}{2}\right] + 7\left[\tbinom{n}{3} - \tbinom{0}{3}\right] + 12\left[\tbinom{n}{4} - \tbinom{0}{4}\right] + 6\left[\tbinom{n}{5} - \tbinom{0}{5}\right],$$
$$\Sigma^1 n^3 = 8\left[\tbinom{n-1}{2} - \tbinom{-1}{2}\right] + 19\left[\tbinom{n-1}{3} - \tbinom{-1}{3}\right] + 18\left[\tbinom{n-1}{4} - \tbinom{-1}{4}\right] + 6\left[\tbinom{n-1}{5} - \tbinom{-1}{5}\right],$$
$$\Sigma^1 n^3 = 27\left[\tbinom{n-2}{2} - \tbinom{-2}{2}\right] + 37\left[\tbinom{n-2}{3} - \tbinom{-2}{3}\right] + 24\left[\tbinom{n-2}{4} - \tbinom{-2}{4}\right] + 6\left[\tbinom{n-2}{5} - \tbinom{-2}{5}\right].$$

The preceding example contains binomial coefficients with negative upper indices. These coefficients are determined by the identity $\binom{-n}{k} = (-1)^k \binom{k+n-1}{k}$. Setting $t = 0$ in Proposition (3.1) gives another well-known identity for sums of powers, discussed in [5, p. 190] and in [8].

Corollary 3.3. *For non-negative integers n and m*

$$\Sigma^1 n^m = \sum_{j=0}^{m} \binom{n+1}{j+1} \Delta^j 0^m.$$

The following example illustrates the identity.

Example 3.4. *For non-negative integer n, we have*

$$\begin{aligned}
\Sigma^1 n^1 &= 1\tbinom{n+1}{1},\\
\Sigma^1 n^1 &= 0\tbinom{n+1}{1} + 1\tbinom{n+1}{2},\\
\Sigma^1 n^2 &= 0\tbinom{n+1}{1} + 1\tbinom{n+1}{2} + 2\tbinom{n+1}{3},\\
\Sigma^1 n^3 &= 0\tbinom{n+1}{1} + 1\tbinom{n+1}{2} + 6\tbinom{n+1}{3} + 6\tbinom{n+1}{4},\\
\Sigma^1 n^4 &= 0\tbinom{n+1}{1} + 1\tbinom{n+1}{2} + 14\tbinom{n+1}{3} + 36\tbinom{n+1}{4} + 24\tbinom{n+1}{5},\\
\Sigma^1 n^5 &= 0\tbinom{n+1}{1} + 1\tbinom{n+1}{2} + 30\tbinom{n+1}{3} + 150\tbinom{n+1}{4} + 240\tbinom{n+1}{5} + 120\tbinom{n+1}{6}.
\end{aligned}$$

The coefficients $0, 1, 2, 0, 1, 6, 6, 0, 1, 14, 36, 24, \ldots$ are given by the Stirling numbers of the second kind through $a_k = k!\left\{{n \atop k}\right\}$. They form the sequence A131689 in the OEIS [9]. Thus, Stirling numbers of the second kind $\left\{{n \atop k}\right\}$ arise naturally from forward Newton interpolation for powers. Setting $t = 1$ in Proposition (3.1) gives another well-known special case.

Corollary 3.5. *For non-negative integers n and m,*

$$\Sigma^1 n^m = \sum_{j=0}^{m} \binom{n}{j+1} \Delta^j 1^m.$$

This identity is discussed, for example, in [10]. The following example illustrates it.

Example 3.6. *For non-negative integer n, we have*

$$\Sigma^1 n^0 = 1\binom{n}{1},$$

$$\Sigma^1 n^1 = 1\binom{n}{1} + 1\binom{n}{2},$$

$$\Sigma^1 n^2 = 1\binom{n}{1} + 3\binom{n}{2} + 2\binom{n}{3},$$

$$\Sigma^1 n^3 = 1\binom{n}{1} + 7\binom{n}{2} + 12\binom{n}{3} + 6\binom{n}{4},$$

$$\Sigma^1 n^4 = 1\binom{n}{1} + 15\binom{n}{2} + 50\binom{n}{3} + 60\binom{n}{4} + 24\binom{n}{5},$$

$$\Sigma^1 n^5 = 1\binom{n}{1} + 31\binom{n}{2} + 180\binom{n}{3} + 390\binom{n}{4} + 360\binom{n}{5} + 120\binom{n}{6}.$$

The displayed coefficients belong to OEIS sequence A028246 [9]. The work in [10] gives formulas for sums of powers in terms of generalized Stirling numbers of the second kind $\left\{ {k \atop j} \right\}_r$. Specifically, it gives the following result.

Proposition 3.7 (Cereceda (2022))**.** *For non-negative integers n and k,*

$$\Sigma^1 n^k = \sum_{j=0}^{k} j! \left[\binom{n+1-r}{j+1} + (-1)^j \binom{r+j-1}{j+1} \right] \left\{ {k \atop j} \right\}_r. \tag{1}$$

The same work also gives

$$\Sigma^1 n^k = \sum_{j=0}^{k} j! \left[\binom{n+1+r}{j+1} + \binom{r+1}{j+1} \right] \left\{ {k \atop j} \right\}_{-r}. \tag{2}$$

Equation (1) is a special case of Proposition (3.1), obtained by setting $t = r$ and applying the upper-binomial negation identity in Lemma (1.8) to $\binom{1-t}{k+1}$. The forward finite differences of powers can also be expressed in terms of generalized Stirling numbers of the second kind:

$$\Delta^k t^m = k! \left\{ {m \atop k} \right\}_t. \tag{3}$$

This identity shows that generalized Stirling numbers of the second kind $\left\{ {n \atop k} \right\}_t$ arise naturally from forward Newton interpolation for powers evaluated at an arbitrary integer t. Equation (2) is another special case of Proposition (3.1), obtained by setting $t = -r$ and using Equation (3). Setting $t = 4$ in Proposition (3.1) gives the following less familiar formulas for sums of powers.

Example 3.8. *For non-negative integer n, we have*

$$\Sigma^1 n^0 = 1\left(\tbinom{n-3}{1} + \tbinom{3}{1}\right),$$
$$\Sigma^1 n^1 = 4\left(\tbinom{n-3}{1} + \tbinom{3}{1}\right) + 1\left(\tbinom{n-3}{2} - \tbinom{4}{2}\right),$$
$$\Sigma^1 n^2 = 16\left(\tbinom{n-3}{1} + \tbinom{3}{1}\right) + 9\left(\tbinom{n-3}{2} - \tbinom{4}{2}\right) + 2\left(\tbinom{n-2}{3} + \tbinom{5}{3}\right),$$
$$\Sigma^1 n^3 = 64\left(\tbinom{n-3}{1} + \tbinom{3}{1}\right) + 61\left(\tbinom{n-3}{2} - \tbinom{4}{2}\right) + 30\left(\tbinom{n-3}{3} + \tbinom{5}{3}\right) + 6\left(\tbinom{n-3}{4} - \tbinom{6}{4}\right).$$

The coefficients $1, 4, 1, 16, 9, \ldots$ form the sequence A391633 in the OEIS [9]. We now consider double sums of powers. Setting $r = 1$ in the hockey-stick identity from Lemma (1.6) and including the correction terms $\Sigma^r n^0$ gives the following formula.

Proposition 3.9 (Double sums of powers)**.** *For non-negative integers n and m and an arbitrary integer t,*

$$\Sigma^2 n^m = \sum_{k=0}^{m} \left[\binom{n-t+2}{k+2} - \binom{1-t}{k+1}\Sigma^1 n^0 - \binom{2-t}{k+2}\Sigma^0 n^0\right] \Delta^k t^m.$$

The zero-power sums satisfy $\Sigma^1 n^0 = n$ and $\Sigma^0 n^0 = 1$. Applying the upper-binomial negation identity from Lemma (1.8) gives the following equivalent formula.

Proposition 3.10 (Double sums of powers negated)**.** *For non-negative integers n and m and an arbitrary integer t,*

$$\Sigma^2 n^m = \sum_{k=0}^{m} \left[\binom{n-t+2}{k+2} + (-1)^k \binom{k+t-1}{k+1} n + (-1)^{k+1} \binom{k+t-1}{k+2}\right] \Delta^j t^m.$$

Setting $t = 5$ in Proposition (3.10) gives the following example.

Example 3.11. *For non-negative integer n, we have*

$$\begin{aligned}
\Sigma^2 n^0 &= 1\left(\tbinom{n-3}{2} + \tbinom{4}{1}n - \tbinom{4}{2}\right),\\
\Sigma^2 n^1 &= 5\left(\tbinom{n-3}{2} + \tbinom{4}{1}n - \tbinom{4}{2}\right) + 1\left(\tbinom{n-3}{3} - \tbinom{5}{2}n + \tbinom{5}{3}\right),\\
\Sigma^2 n^2 &= 25\left(\tbinom{n-3}{2} + \tbinom{4}{1}n - \tbinom{4}{2}\right) + 11\left(\tbinom{n-3}{3} - \tbinom{5}{2}n + \tbinom{5}{3}\right) + 2\left(\tbinom{n-3}{4} + \tbinom{6}{3}n - \tbinom{6}{4}\right),\\
\Sigma^2 n^3 &= 125\left(\tbinom{n-3}{2} + \tbinom{4}{1}n - \tbinom{4}{2}\right) + 91\left(\tbinom{n-3}{3} - \tbinom{5}{2}n + \tbinom{5}{3}\right) + 36\left(\tbinom{n-3}{4} + \tbinom{6}{3}n - \tbinom{6}{4}\right)\\
&\quad + 6\left(\tbinom{n-3}{5} - \tbinom{7}{3}n + \tbinom{7}{4}\right).
\end{aligned}$$

The coefficients $1, 5, 1, 25, 11, 2, \ldots$ form the sequence A391635 in the OEIS [9]. Repeating the procedure used for $\Sigma^1 n^m$ and $\Sigma^2 n^m$ gives a general formula for r-fold sums of powers.

Theorem 3.12 (Multifold sums of powers)**.** *For non-negative integers r, n, and m, and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^{m}\left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1}\binom{r-s-t}{k+r-s}\Sigma^s n^0\right]\Delta^k t^m.$$

Using Lemma (1.7), we obtain the following pure binomial form.

Proposition 3.13 (Multifold binomial sums of powers)**.** *For non-negative integers r, n, and m, and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^{m}\left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1}\binom{r-s-t}{k+r-s}\binom{s+n-1}{s}\right]\Delta^k t^m.$$

Explicitly,

$$\Sigma^r n^m = \sum_{k=0}^{m}\left[\tbinom{n-t+r}{k+r} - \tbinom{r-t}{k+r}\tbinom{n-1}{0} - \tbinom{r-1-t}{k+r-1}\tbinom{n}{1} - \cdots - \tbinom{2-t}{k+2}\tbinom{r+n-3}{r-2} - \tbinom{1-t}{k+1}\tbinom{r+n-2}{r-1}\right]\Delta^k t^m.$$

The following example illustrates these formulas.

Example 3.14. *For non-negative integers r, n, and m we have*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n+r-0}{k+r} - \sum_{s=0}^{r-1} \binom{r-s}{k+r-s} \binom{s+n-1}{s} \right] \Delta^k 0^m,$$

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n+r-1}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-1}{k+r-s} \binom{s+n-1}{s} \right] \Delta^k 1^m,$$

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n+r-2}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-2}{k+r-s} \binom{s+n-1}{s} \right] \Delta^k 2^m,$$

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n+r-3}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-3}{k+r-s} \binom{s+n-1}{s} \right] \Delta^k 3^m.$$

These formulas are polynomials in n with integer coefficients given by forward finite differences of powers evaluated at an arbitrary integer t. The following example lists the cases $0 \le t \le 4$.

Example 3.15. *For $r = 1$ and $m = 3$, Proposition 3.13 yields*

$$t = 0: \quad \Sigma^1 n^3 = n \cdot 0 + \tfrac{1}{2}(n + n^2) \cdot 1 + \tfrac{1}{6}(-n + n^3) \cdot 6 + \tfrac{1}{24}(2n - n^2 - 2n^3 + n^4) \cdot 6,$$

$$t = 1: \quad \Sigma^1 n^3 = n \cdot 1 + \tfrac{1}{2}(-n + n^2) \cdot 7 + \tfrac{1}{6}(2n - 3n^2 + n^3) \cdot 12 + \tfrac{1}{24}(-6n + 11n^2 - 6n^3 + n^4) \cdot 6,$$

$$t = 2: \quad \Sigma^1 n^3 = n \cdot 8 + \tfrac{1}{2}(-3n + n^2) \cdot 19 + \tfrac{1}{6}(11n - 6n^2 + n^3) \cdot 18 + \tfrac{1}{24}(-50n + 35n^2 - 10n^3 + n^4) \cdot 6,$$

$$t = 3: \quad \Sigma^1 n^3 = n \cdot 27 + \tfrac{1}{2}(-5n + n^2) \cdot 37 + \tfrac{1}{6}(26n - 9n^2 + n^3) \cdot 24 + \tfrac{1}{24}(-154n + 71n^2 - 14n^3 + n^4) \cdot 6,$$

$$t = 4: \quad \Sigma^1 n^3 = n \cdot 64 + \tfrac{1}{2}(-7n + n^2) \cdot 61 + \tfrac{1}{6}(47n - 12n^2 + n^3) \cdot 30 + \tfrac{1}{24}(-342n + 119n^2 - 18n^3 + n^4) \cdot 6.$$

The coefficients have the following forms.

- For $t = 0$, the coefficients $0, 1, 6, 6$ belong to the OEIS sequence A131689.
- For $t = 1$, the coefficients $1, 7, 12, 6$ belong to the OEIS sequence A028246.
- For $t = 2$, the coefficients $8, 19, 18, 6$ belong to the OEIS sequence A038719.

- For $t = 3$, the coefficients $27, 37, 24, 6$ belong to the OEIS sequence A391552.
- For $t = 4$, the coefficients $64, 61, 30, 6$ belong to the OEIS sequence A391633.

4. Stirling Form

We first express forward finite differences in terms of Stirling numbers.

Lemma 4.1 (Stirling finite differences)**.** *For non-negative integers k and m and an arbitrary integer t,*

$$\Delta^k t^m = \sum_{j=0}^{m} \left\{ {m \atop k+j} \right\} \binom{t}{j} (k+j)!.$$

Proof. Applying the well-known identity $t^m = \sum_{j=0}^{m} \left\{ {m \atop j} \right\} \binom{t}{j} j!$ to the forward difference gives the stated formula. □

Substituting this expression into Theorem (3.12) gives the following formula.

Proposition 4.2 (Multifold Stirling sums)**.** *For non-negative integers r, n, and m and an arbitrary integer t,*

$$\Sigma^r n^m = \sum_{k=0}^{m} \sum_{j=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \Sigma^s n^0 \right] \left\{ {m \atop k+j} \right\} \binom{t}{j} (k+j)!.$$

Applying Lemma (1.7) gives the following pure binomial form.

Proposition 4.3 (Multifold Stirling-Binomial sums)**.** *For non-negative integers r, n, and m and an arbitrary integer t,*

$$\Sigma^r n^m = \sum_{k=0}^{m} \sum_{j=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \binom{s+n-1}{s} \right] \left\{ {m \atop k+j} \right\} \binom{t}{j} (k+j)!.$$

5. Eulerian Form

We begin with Worpitzky's identity.

Lemma 5.1 (Worpitzky Identity)**.** *For non-negative integers t and m,*

$$t^m = \sum_{k=0}^{m} \binom{t+k}{m} \left\langle {m \atop k} \right\rangle,$$

where $\left\langle {n \atop k} \right\rangle$ *are Eulerian numbers.*

Proof. The identity is given by J. Worpitzky in [11]. □

The following lemma expresses finite differences of powers in terms of Eulerian numbers $\left\langle {m \atop s} \right\rangle$.

Lemma 5.2 (Eulerian finite differences)**.** *For non-negative integers* k *and* m *and an arbitrary integer* t*,*

$$\Delta^k t^m = \sum_{j=0}^{m} \binom{t+j}{m-k} \left\langle {m \atop j} \right\rangle.$$

Proof. This follows from the binomial recurrence $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$ and Worpitzky's identity in Lemma (5.1). □

Substituting this expression into Theorem (3.12) gives the following formula.

Proposition 5.3 (Multifold Eulerian sums)**.** *For non-negative integers* r*,* n*, and* m *and an arbitrary integer* t*,*

$$\Sigma^r n^m = \sum_{k=0}^{m} \sum_{j=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \Sigma^s n^0 \right] \binom{t+j}{m-k} \left\langle {m \atop j} \right\rangle.$$

Applying Lemma (1.7) gives the following pure binomial form.

Proposition 5.4 (Multifold Eulerian-Binomial sums)**.** *For non-negative integers* r*,* n*, and* m *and an arbitrary integer* t*,*

$$\Sigma^r n^m = \sum_{k=0}^{m} \sum_{j=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \binom{s+n-1}{s} \right] \binom{t+j}{m-k} \left\langle {m \atop j} \right\rangle.$$

6. Backward Difference Form

We first relate forward and backward finite differences.

Lemma 6.1 (Forward-Backward relation)**.** *For non-negative integers* k *and* m *and an arbitrary integer* t*,*

$$\Delta^k t^m = \nabla^k (t+k)^m,$$

where

$$\nabla^k t^m = \sum_{j=0}^{k} (-1)^j \binom{k}{j} (t-j)^m.$$

Applying Lemma (6.1) to Theorem (3.12) gives the following formula.

Proposition 6.2 (Multifold Backward sums)**.** *For non-negative integers r, n, and m and an arbitrary integer t,*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \Sigma^s n^0 \right] \nabla^k (t+k)^m.$$

Applying Lemma (1.7) gives the following pure binomial form.

Proposition 6.3 (Multifold Backward-Binomial sums)**.** *For non-negative integers r, n, and m and an arbitrary integer t,*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \binom{s+n-1}{s} \right] \nabla^k (t+k)^m.$$

7. Central Difference Form

We first relate forward and central finite differences.

Lemma 7.1 (Forward-central relation)**.** *For non-negative integers k and m and an arbitrary integer t,*

$$\Delta^k t^m = \delta^k \left(t + \frac{k}{2} \right)^m,$$

where

$$\delta^k t^m = \sum_{j=0}^{k} (-1)^j \binom{k}{j} \left(t + \tfrac{k}{2} - j \right)^m.$$

Applying Lemma (7.1) to Theorem (3.12) gives the following formula.

Proposition 7.2 (Multifold Central sums)**.** *For non-negative integers r, n, and m and an arbitrary integer t,*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \Sigma^s n^0 \right] \delta^k \left(t + \frac{k}{2} \right)^m.$$

Applying Lemma (1.7) gives the following pure binomial form.

Proposition 7.3 (Multifold Central-Binomial sums)**.** *For non-negative integers r, n, and m and an arbitrary integer t,*

$$\Sigma^r\, n^m = \sum_{k=0}^{m} \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \binom{s+n-1}{s} \right] \delta^k \left(t + \frac{k}{2} \right)^m .$$

8. Conclusions

In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the forward Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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Mathematica Programs

Use the Mathematica programs in the project repository to validate the results. Table 1 lists the programs and the results they validate. The repository is available at GitHub.

Mathematica Function	Validates or Prints
`ValidateMultifoldSumsOfPowers.txt`	Validates Theorem (3.12)
`ValidateMultifoldBinomialSumsOfPowers.txt`	Validates Proposition (3.13)
`ValidateMultifoldStirlingSums.txt`	Validates Proposition (4.2)
`ValidateMultifoldStirlingBinomialSums.txt`	Validates Proposition (4.3)
`ValidateMultifoldEulerianSums.txt`	Validates Proposition (5.3)
`ValidateMultifoldEulerianBinomialSums.txt`	Validates Proposition (5.4)
`ValidateMultifoldBackwardSums.txt`	Validates Proposition (6.2)
`ValidateMultifoldBackwardBinomialSums.txt`	Validates Proposition (6.3)
`ValidateMultifoldCentralSums.txt`	Validates Proposition (7.2)
`ValidateMultifoldCentralBinomialSums.txt`	Validates Proposition (7.3)

Table 1. Mathematica programs and the results they validate.

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- **Sources:** github.com/kolosovpetro/NewtonsFormulaAndSumsOfPowers

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